TARUN SINGH

PhD Scholar | Department of Civil Engineering | Indian Institute of Technology Roorkee, India
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RESEARCH PROFILE

PhD researcher working on the constitutive modelling of rate-dependent materials, with a focus on the finite-deformation behaviour of viscoelastic solids. My research involves rheological connections, evolving natural configurations, and thermodynamically consistent constitutive formulations. It combines theoretical developments in continuum mechanics with analytical solutions, numerical integration, material parameter identification, and computational implementation.

RESEARCH INTERESTS

Finite-deformation continuum mechanics and constitutive theory • Finite viscoelasticity and viscoplasticity • Evolving natural configurations and rheological connections • Geometric perspectives in continuum mechanics • Anisotropic constitutive modelling of rate-dependent materials • Thermodynamics and numerical implementation of constitutive models

EDUCATION

PhD in Civil Engineering | 2023–Present
Indian Institute of Technology Roorkee, India
Research: Constitutive modelling of rate-dependent materials | Supervisor: Dr. Sandipan Paul

M.Tech. in Structural and Construction Engineering | 2020–2022
Dr. B. R. Ambedkar National Institute of Technology Jalandhar, India
Thesis: Carbonation resistance of concrete made with iron slag and recycled concrete aggregate

B.Tech. in Civil Engineering | 2015–2019
Rewa Engineering College, Rewa, Madhya Pradesh, India

RESEARCH EXPERIENCE AND CONTRIBUTIONS

- Developed finite-deformation constitutive frameworks for rate-dependent materials, with a focus on viscoelastic solids.
- Extended rheological connections to finite deformations using multiple natural configurations.
- Formulated Kelvin–Voigt-type viscoelastic constitutive models using the theory of evolving natural configurations.
- Developed a deformation-mode-specific constitutive framework based on scalar, conjugate stress/strain pairs and applied it to soft-material characterization.
- Performed numerical integration of nonlinear constitutive equations and parameter identification using experimental response data.

JOURNAL PUBLICATIONS – DOCTORAL RESEARCH

1. **T. Singh** and S. Paul (2027). *"A deformation mode-specific constitutive framework for finite viscoelasticity based on scalar, conjugate stress/strain pairs: Applications to characterization of soft materials."* Journal of the Mechanics and Physics of Solids, 219, Article 106895. DOI
2. **T. Singh** and S. Paul (2026). *"Constitutive modeling of Kelvin–Voigt type viscoelastic solids undergoing finite deformations using the theory of evolving natural configurations."* International Journal of Solids and Structures, 341, Article 114336. DOI
3. **T. Singh** and S. Paul (2025). *"On the extension of the concept of rheological connections to a finite deformation framework using multiple natural configurations."* International Journal of Engineering Science, 212, Article 104267. DOI

EARLIER PUBLICATIONS – M.TECH. RESEARCH

1. N. Singh, A. Singh, N. Ankur, P. Kumar, M. Kumar, and **T. Singh** (2022). *"Reviewing the properties of recycled concrete aggregates and iron slag in concrete."* Journal of Building Engineering, 60, 105150.
2. N. Singh, N. Ankur, **T. Singh**, M. Kumar, A. Singh, and P. Kumar (2022). *"Non-destructive evaluation of iron slag concrete containing recycled concrete aggregates."* Indian Concrete Journal, 96(11).
3. N. Singh, **T. Singh**, M. Kumar, A. Singh, and P. Kumar (2022). *"Investigating the fresh state performance of concrete containing iron slag and recycled concrete aggregates."* Materials Today: Proceedings, 65, 1467–1477.

CONFERENCE PRESENTATIONS

- T. Singh and S. Paul (2026). "Constitutive modeling of viscoelastic solids undergoing finite deformations using the theory of evolving natural configurations." IUTAM Symposium: Rate Dependent Mechanics and Multiscale Modelling of Materials, IIT Delhi.
- T. Singh and S. Paul (2026). "Thermodynamically consistent constitutive model for viscoelastic solids under finite deformation using evolving natural configurations." 7th Indian Conference on Applied Mechanics, IIT Kanpur.
- T. Singh and S. Paul (2025). "Challenges in extending rheological connections to a finite deformation theory." 14th International Symposium on Plasticity and Impact Mechanics (IMPLAST 2025), IIT Roorkee.

COMPUTATIONAL AND TECHNICAL SKILLS

Programming: MATLAB, Fortran

Finite-element analysis: Abaqus/Standard

Constitutive implementation: Finite-strain viscoelastic UMAT development and verification

Numerical methods: Numerical integration of constitutive equations; parameter identification and model calibration

COMPUTATIONAL IMPLEMENTATION AND ONGOING RESEARCH

- Implemented finite-strain viscoelastic constitutive response in an Abaqus/Standard UMAT using Fortran.
- Current research directions include extensions towards viscoplasticity and anisotropic formulations for other material symmetry classes.